



LUNDS
UNIVERSITET

Tentamensskrivning
Topologi
Lördag den 26 april 2025
Skrivtid: 08.00 – 13:00

Matematikcentrum

Matematik NF

Inga hjälpmedel. Använd institutionens papper, skriv bara på den ena sidan och högst en uppgift på varje papper. Fyll i omslaget fullständigt och skriv nummret på varje ark. Skriv tydligt. Ge klara och kortfattade motiveringar, rita gärna figur i förekommande fall.

No books, notes, computational devices etc. are allowed. Use paper supplied by the Department, write only on one side of each paper, and treat at most one exercise on each paper. Use clear handwriting and give clear careful motivations, including sketches where appropriate. Fill in the form completely and write your number on each sheet of paper.

1. Let $\mathbb{R}^3$ be equipped with the standard topology induced by the Euclidean metric and let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the function given by

$$f(x, y, z) = \log(|xyz| + 1).$$

Put $A = f^{-1}(\{0\})$, $B = f^{-1}((0, 1))$, $C = A \cap (\mathbb{R} \times [1, 2] \times [1, 2])$.

Prove whether the following statements are true or false.

- a) A is compact, 1,5
- b) A is path-connected, 1,5
- c) B is connected, 1,5
- d) C is compact. 1,5

2. Let $X = [-1, 1]$.

- a) Show that

$$\mathcal{S} = \{(a, 1] : -1 \leq a < 0\} \cup \{[-1, b) : 1 \geq b > 0\}$$

2

is a subbasis of a topology on X .

- b) Show that

2

$$\mathcal{T} = \{(a, 1] : -1 \leq a < 0\} \cup \{[-1, b) : 1 \geq b > 0\} \cup \{(a, b) : -1 \leq a < 0, 1 \geq b > 0\} \cup \{X, \emptyset\}$$

is the topology generated by $\mathcal{S}$. $\mathcal{T}$ is called the *overlapping interval topology*.

- c) Show that $(X, \mathcal{T})$ is T_0 , that is, for any $x, y \in X$ with $x \neq y$, there exists $U \in \mathcal{T}$ with $x \in U$, $y \notin U$ or $y \in U$, $x \notin U$. 1
- d) Show that $(X, \mathcal{T})$ is not T_1 . Recall that a topology is T_1 , if all one-point sets are closed. 1

3. Let $(X, \mathcal{T})$ be the topological space from Question 2. You may use all parts of Question 2.

- a) Describe all sequences $(x_k)_{k \in \mathbb{N}}$ in X such that $(x_k) \xrightarrow{k \rightarrow \infty} 0$ with respect to $\mathcal{T}$. 1,5

Var god vänd!

- b) Describe all sequences $(x_k)_{k \in \mathbb{N}}$ in X such that $(x_k) \xrightarrow[k \rightarrow \infty]{} 1$ with respect to $\mathcal{T}$. 1,5
- c) Describe all sequences $(x_k)_{k \in \mathbb{N}}$ in X such that $(x_k) \xrightarrow[k \rightarrow \infty]{} 1/2$ with respect to $\mathcal{T}$. 1,5
- d) Conclude that $(X, \mathcal{T})$ is sequentially compact. 1,5
4. Let $(X, \mathcal{T})$ be a connected compact Hausdorff space with at least two elements. Show that X has uncountably many elements. 6
5. We consider $\mathbb{R}$ with the standard topology induced by the Euclidean metric. For $n \in \mathbb{N}$, let $f_n : \mathbb{R} \rightarrow \mathbb{R}$, $f_n(x) = n^3 \sinh(x/n) - n^2 x$.
- a) Show that the sequence of functions (f_n) converges pointwise on $\mathbb{R}$, and determine the limit function f . 2
- b) Show that the family $\{f_n : n \in \mathbb{N}\}$ is pointwise bounded and equicontinuous on $\mathbb{R}$. 2
- c) Show that the sequence of functions (f_n) converges uniformly to the limit function f on $[-N, N]$ for each $N \in \mathbb{N}$, but that it does not converge uniformly on $\mathbb{R}$. 2